

Title

Adversarially Asymmetric Evidence: A Calibrated Likelihood-Ratio Model for Fake News Detection

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1. Abstract

Fake news detectors score articles on features the author of the article chooses. Headline wording, emotional register, the presence of named sources, the absence of clickbait — every one of these is set by whoever wrote the text, and a detector that rewards them is a detector a competent fabricator can satisfy on purpose. We measured this on our own earlier system: invented articles written in plain newsroom register scored 7.3 out of 10, and four of six deceptive items were presented to the reader as credible.

The underlying error is treating misinformation detection as ordinary classification. In ordinary classification the features are drawn from nature. Here an adversary controls them, and that changes what a feature is worth. We formalise the consequence as an asymmetry: **an observation under the author's control may incriminate an article but can never exonerate it**, because a capable adversary can always produce the exonerating value. Detectors trained on corpora of crude fabrications learn the opposite and do not notice, since such corpora contain no competent fakes to contradict them.

The scoring model follows from this. Each observation contributes a likelihood ratio — how much more often it occurs in fabricated than in genuine articles — estimated from labelled data rather than assigned by hand, and accumulated as additive log-odds in decibans. The output is $P(\text{fake news})$: a probability, which unlike a 0–10 score refers to something outside the system and can be checked against outcomes. Three properties follow directly from the arithmetic rather than from special cases in the code. An uninformative observation has a ratio of one and contributes exactly nothing. An observation nobody has measured contributes nothing, instead of a number someone invented. And author-controlled evidence is admitted one-sidedly, at every stage — including after calibration, where we found the correction silently reversing the rule and restoring the exculpatory weight it was meant to remove.

Measured on held-out ISOT data, with likelihood ratios estimated on a training half only: accuracy rises from 0.888 to 0.954 against the hand-weighted rule this replaces, expected calibration error where the system accuses falls from 0.229 to 0.127, and precision among flagged articles rises from 0.848 to 0.975. Isotonic calibration reduces overall calibration error from 0.335 to 0.056 on the content model. Two further results matter more than the headline figures. The four content checks correlate at $r = 0.78$ and constitute **1.2 effective independent factors of four**, so the previous design counted one piece of evidence four times. And every residual calibration error in the shipped system points in the direction of greater suspicion: the model never overstates confidence in an article's favour.

On an adversarial set of fabrications written to read like ordinary reporting, the old rule presented four of six as credible and the current model presents none, while condemning none of the genuine articles. Corroboration is counted in independent sources rather than articles — collapsing shared ownership, wire syndication and near-duplicate reprints — and a deterministic rule set issues a verdict with its grounds. Language is identified locally across fifteen languages, claims are searched across language boundaries, and video, screenshot and forwarded text are ingested with provenance. Every decisive step is deterministic local code: the test suite and both benchmarks run with all external keys removed.

Keywords — fake news detection, adversarial evidence, likelihood ratio, weight of evidence, probability calibration, source independence, multilingual fact verification

2. Base Paper

S. Amri, H.-C. Mputu Boleilanga, and E. Aïmeur, "ExFake: Towards an Explainable Fake News Detection Based on Content and Social Context Information," arXiv:2311.10784, Nov. 2023.

ExFake judges an online post by fusing content, social-context signals (the credibility and history of the users circulating it) and evidence from trusted external entities, coupling the decision to an explainable-AI assistant for ordinary readers. It matches this project on objective, on method — fusion of heterogeneous signal families rather than one classifier — on external grounding, and on audience.

Its fusion is additive and symmetric, which is the property we identify as unsound under adversarial control and replace.

Alternatives considered and rejected as the base. *VeraCT Scan* [3] (2024) already occupies the retrieval-and-justify ground, leaving a narrower delta. *TELLER* [2] (Findings of ACL 2024) explains through logic rules rather than retrieved evidence. *TRUST-VL* [6] (EMNLP 2025) is a vision-language model trained on 198K samples; adopting it as the base would frame this project as an under-resourced attempt at the same thing, whereas the contribution here is in how evidence is weighed. Every comparative reference in Section 8 is from 2023 or later; the two pre-2020 entries are cited only as the provenance of standard techniques.

3. The Defect

Additive scoring over author-controlled features fails in a direction that is easy to miss and hard to repair by tuning.

Observation	Set by the author?	Usable as exculpatory evidence?
Headline style	Yes	No
Neutral wording	Yes	No
Absence of persuasion techniques	Yes	No
Named sources, attributed quotes	Yes — they can be invented	No
Publisher's historical record	No — earned over years	Yes
Independent corroboration	No — requires other newsrooms	Yes

Only the last two rows resist the author. The previous design weighted all of them together and, when corroboration found nothing, redistributed its share across the rest — so an absence of evidence increased the influence of precisely the signals a fabricator controls. Measured on the adversarial set, this produced a mean score of 7.3/10 for invented articles.

No choice of weights repairs it, because the fault is not in the magnitudes but in admitting the evidence in both directions at all.

ISOT cannot expose this. Its fabrications are crude, so on that corpus polished writing genuinely does predict a genuine article, and a detector that learns the association scores well while being wide open to a competent fake. This is why the project carries a second benchmark.

4. Points of Novelty

All six are implemented, tested and measured.

N1 · Adversarial evidence asymmetry. An observation the author controls is admitted only in the incriminating direction: its likelihood ratio is clamped at one on the exculpatory side, at every stage of the computation. This is the central claim, and it is not a heuristic — it follows from the observation that a capable adversary drives $P(\text{favourable} \mid \text{fabricated})$ toward $P(\text{favourable} \mid \text{genuine})$, so the exculpatory ratio tends to one whatever a crude-fake corpus says. We know of no fake-news system that applies it. The rule also has to survive calibration: fitting a correction on labelled data silently reversed it and handed back a large exculpatory weight, which our regression test now prevents.

N2 · Measured weights instead of chosen ones. Every ratio is estimated from labelled articles by `eval/estimateWeights.js`, stored with the counts that produced it, shrunk to the conservative end of a 95% interval, and reported to the reader in decibans. "Why 0.18?" is no longer a question the system has to dodge. A factor with too little data to estimate — transparency, at $n=18$ — contributes exactly zero rather than an invented weight.

N3 · Correlation-aware combination. The four content checks were treated as four independent factors. Measured, they correlate at $r = 0.78$ and amount to 1.2 effective independent factors, so their summed weight is scaled accordingly. Without this the same evidence was counted four times.

N4 · An output that means something. $P(\text{fake news})$ rather than a 0–10 score with no referent. It is checkable — across articles scored 0.7, about seventy per cent should be fabricated — and we check it, reporting calibration error separately for the accusing and exonerating directions, because a single figure conceals the trade the asymmetry makes.

N5 · Independence-aware corroboration. Counting articles overstates corroboration, often tenfold. A curated ownership model of 70 media groups, agency-syndication detection and near-duplicate collapse reduce retrieved coverage to the sources it actually represents, and every merge is reported with its reason. A deterministic relevance gate additionally discards retrieved coverage that shares no anchor — place, name or number — with the claim, after the model has spoken: on the adversarial set the model had reported an invented chemical leak as *supported* by real coverage of an unrelated evacuation.

N6 · The decisive logic is local and deterministic. Likelihood ratios, independence, calibration, verdict rules and language identification contain no model call. The language model extracts claims and judges stance over retrieved text: a sensor, not the judge. The suite (76 tests) and both benchmarks run with every API key removed.

5. What the System Does Not Claim

- **It cannot certify truth.** *Corroborated* means independent outlets report the same facts — the strongest evidence available without primary investigation, and not proof.
- **It abstains, and says so.** *Unverified* is a first-class outcome. Most claims outside major news land there.

- **Its probability is calibrated on ISOT, not in deployment.** The priors and corroboration rates are declared parameters, stated with their justification and reported as declared in every analysis. They are measurable and the scripts exist; until run, the system says which numbers rest on assumption.
- **It does not perform synthetic-media forensics.** A separate problem; the architecture admits such a detector as another factor.

6. Evaluation

Likelihood ratios are estimated on a training half; every figure comes from the held-out half. The split is a hash of the article id, so runs are reproducible.

6.1 Against the rule it replaces

$n = 829$ held-out ISOT articles, content factors only (source identity is hidden on this corpus and historical articles have no live coverage).

	Accuracy	ROC-AUC	ECE when accusing	Precision when accusing
Legacy hand-weighted sum	0.888	0.970	0.229	0.848
Weight of evidence (shipped)	0.954	0.957	0.127	0.975

Calibration of the content model, before and after isotonic regression fitted on the training half: **ECE 0.335 $\rightarrow$ 0.056.**

6.2 The cost of the asymmetry, stated plainly

Turning the clamp off raises ROC-AUC from 0.957 to 0.978 on ISOT. That gain is real and it is a trap: it comes from learning that polished prose indicates a genuine article, which holds on a corpus of crude fakes and fails against a competent one. We report both and ship the clamp.

The clamp also costs calibration on the exonerating side, which is what raises overall ECE from 0.056 to 0.285 in the shipped configuration: the system refuses to become confident that a well-written article is genuine. Every residual error points toward more suspicion, never less — it never overstates confidence in an article's favour.

6.3 Adversarial set

Ten items: five fabrications in plain newsroom register, one high-impact fabrication, two genuine reports, one satire, one commentary. Each scored twice from identical factor outputs.

Measure	Legacy	Current
Deceptive items presented as credible	4 / 6	0 / 6
Verdict matched expectation	—	6 / 6
Genuine articles wrongly condemned	—	0 / 2

Measure	Legacy	Current
Mean legacy score of the fabrications	7.3 / 10	(unchanged — the point)

The fabrications still read well, and always will. What changed is that reading well no longer determines the answer.

6.4 Status of the older ISOT figure

The 94.7% / 82.0% comparison came from a four-factor configuration on a model since withdrawn by its provider. It is retained as the measurement that motivated the architecture, not as a claim about the current system.

7. Reproducing It

```

`bash cd backend npm test                                # 76 tests, no API keys required
node eval/estimateWeights.js                             # measure the likelihood ratios node
eval/validateScoring.js                                  # held-out comparison against the old rule node
eval/validateScoring.js --no-clamp                       # the asymmetry ablation node
eval/adversarialBench.js --no-model                      # well-written fabrications ``

```

8. References

Primary and comparative work (2023–2026)

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Dataset

- [10] H. Ahmed, I. Traore, and S. Saad, "Detection of Online Fake News Using N-Gram Analysis and Machine Learning Techniques," in *Proc. Int. Conf. Intelligent, Secure, and Dependable Systems in Distributed and Cloud Environments*, 2017, pp. 127–138. (*Source of the ISOT corpus, not comparative work.*)

Methodological foundations

- [11] I. J. Good, *Probability and the Weighing of Evidence*. London: Charles Griffin, 1950. (*Weight of evidence in decibans.*)
- [12] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," in *Proc. 34th Int. Conf. Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [13] B. Zadrozny and C. Elkan, "Transforming Classifier Scores into Accurate Multiclass Probability Estimates," in *Proc. 8th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2002, pp. 694–699. (*Isotonic calibration.*)

Appendix A — Short Abstract (150 words)

Fake news detectors score articles on features their authors choose: headline wording, tone, the presence of named sources. A competent fabricator supplies all of them, so such systems reward skilled deception — our earlier pipeline rated invented articles 7.3 out of 10. We identify the cause as treating adversarial detection as ordinary classification, and formalise the correction: an observation under the author's control may incriminate an article but can never exonerate it. Each observation contributes a likelihood ratio estimated from labelled data rather than a hand-chosen weight, accumulated as additive log-odds, yielding a calibrated probability rather than an arbitrary score. On held-out data, accuracy rises from 0.888 to 0.954, precision when accusing from 0.848 to 0.975, and calibration error from 0.229 to 0.127. The four content checks prove to be 1.2 effective independent factors, not four. On well-written fabrications, zero of six now read as credible.